

Trace

A small AI that reads a failed payment and works out what happened to the money — running on the bank's own machines, offline.

Oluwatomiwa Adeaga · deagatech

THE PROBLEM

₦1.07 quadrillion moved. Some of it got stuck.



NIBSS 2024 industry figures; CBN consumer protection report, 2025.

WHY IT IS URGENT

And the clock is set by the regulator

These are mandated ceilings, not internal targets.

Disputed PoS or web transaction



72 hours

Not-on-us ATM failure



48 hours

On-us ATM failure



24 hours

Failed airtime or data · Feb 2026



30 seconds

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days → hours → seconds manual reading does not scale down to seconds

The direction of travel is one way. In February 2026 the CBN and NCC set thirty seconds for failed airtime.

The answer is never in a database column

EXC-2026-0004133**N29,000.00**

customer	Folake Bello · 5928242418
channel	NIP · session 000017082616...
switch	code 00 · Approved or completed successfully
settlement	REF892936787775 · SETTLED
ledger	DR 29,000.00 → beneficiary at Access Bank

remarks

18/08 21:08 CS: cust called again asking for update

19/08 00:14 OPS: refund raised on the switch, awaiting value

Everything above
says this payment
succeeded.

One sentence in the
notes says it did not.

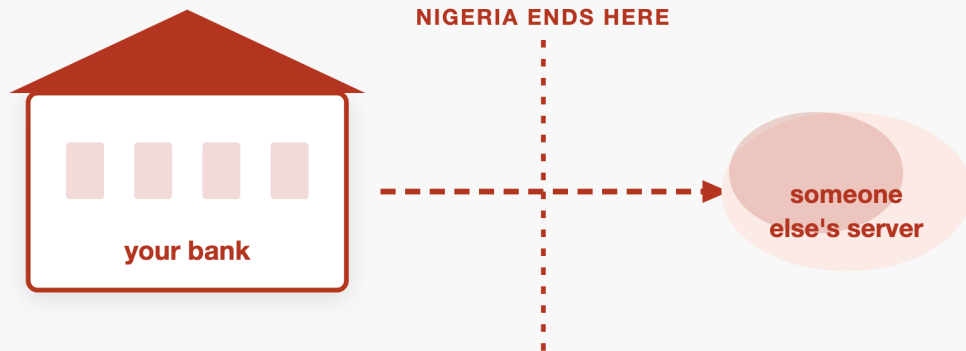
There is no database column
for that sentence. A person
has to read it.

Untrained, this class scores 9%. A rules engine gets 46%. Trained on a bank's own history, 73%.

THE CONSTRAINT EVERYONE HITS

So why not just use ChatGPT?

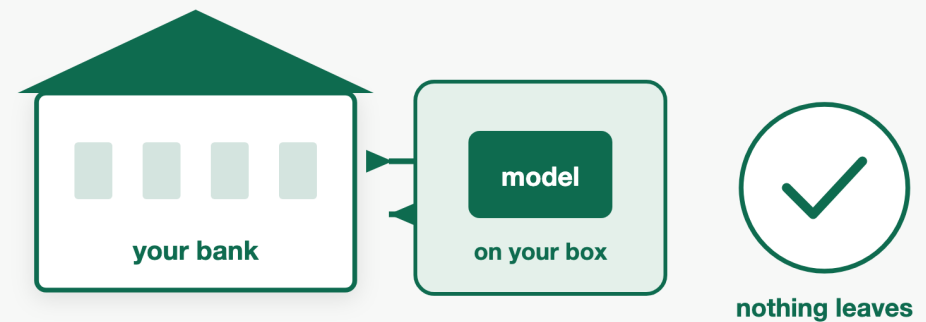
RENTING AN API



Customer names, account numbers and transaction history leave the country.

You own nothing. Price and access can change without you.

RUNNING YOUR OWN



One afternoon of GPU time to train.
Weights, data and test set you keep.

No per-call bill that grows with your own success.

For a Nigerian bank handling customer payment records this is not a preference. It is the difference between a product and a compliance problem.

THE RESULT

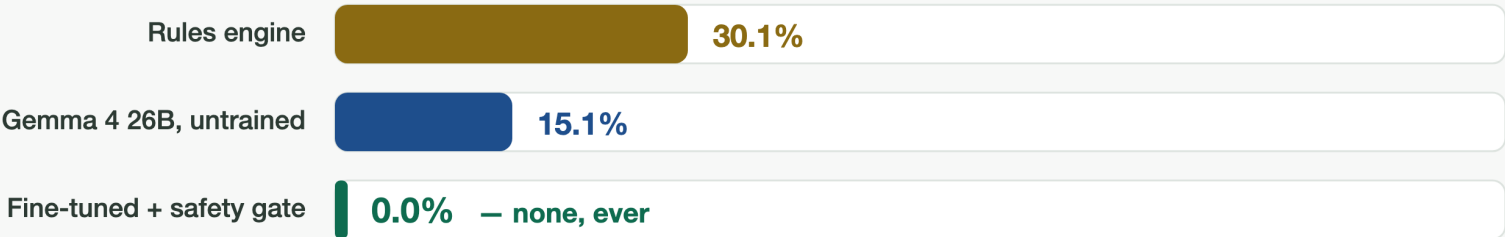
A 2B model beats a 26B model that never saw your data

GETTING THE PROBLEM RIGHT



FRAUD CLOSED WITH NOBODY LOOKING

lower is better

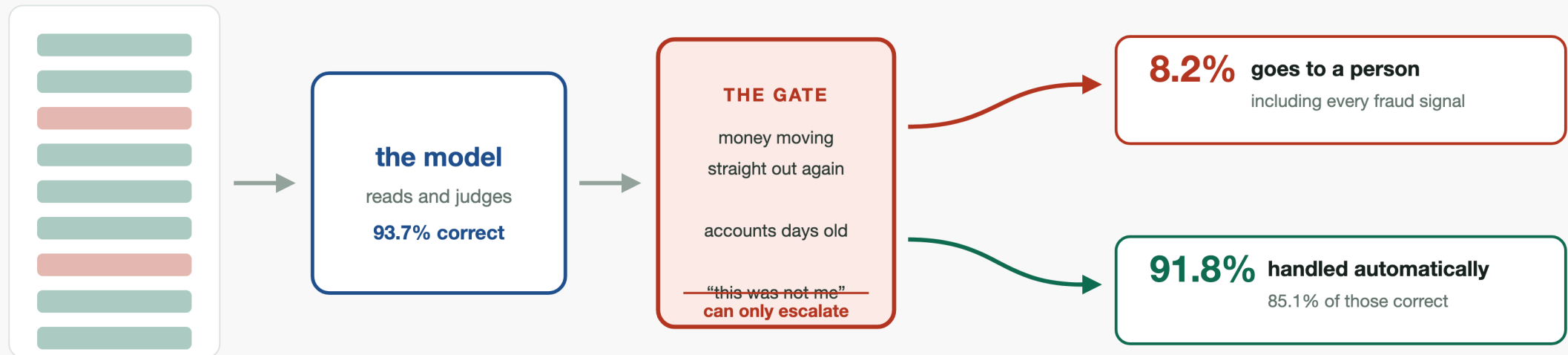


1,000 held-out cases, none seen during training. Every prediction published at github.com/adg-all-day/payment-exceptions-triage

THE SAFETY GATE

It can never quietly close a fraud

EVERY CASE



The model may be wrong. The system may not be dangerous.

Fraud closed with nobody looking: 30.1% under a rules engine, 0.0% here.

WHAT IT TAKES

Built in fourteen hours for about ₦4,500

Not a research project. One rented graphics card, one afternoon of training, and a test set built before the model so the numbers could not flatter it.

Training the model	2 h 42 m	one rented NVIDIA L4
Four evaluation runs	1 h 40 m	5,000 cases scored
Total compute spend	≈ ₦4,500	about \$3, then deleted
The finished model	185 MB	≈2.5 GB packed, runs on a laptop

A bank's exception desk costs more than this in a morning.

What this is not, yet

Everything above is measured on invented data. Here is precisely what has not been proved.

It has not seen one real transaction

Every case is synthetic, generated so the correct answer is certain. That makes the comparison honest, but it does not prove it survives your data.

It has not been tested against drift

The measurement that matters is training on one period and testing on a later one. That needs real history with dates on it.

One class is still weak

Refund-raised-but-never-paid sits at 73%, well below the rest. It is the obvious next thing to fix.

Three datasets were thrown away

Once the clues were too tidy and rules scored 96.6%. Once the evidence was deleted. Once the answers cited evidence that was not there. All recorded in the public build log.

DEAGATECH

The ask

Not prize money.

Anonymised history

Six months of closed exceptions from one break type.
Account numbers hashed, names removed — the text is what matters.

Nothing identifying ever leaves your building.

One person, six weeks

A named contact in operations who can tell us when an answer is wrong and why.

This is the part that cannot be bought.

A number to beat

What your desk resolves per analyst per day today, and how often a case is reopened.

We will publish whether we beat it or not.

You keep the model, the weights, and the test set. If you never speak to us again, it still runs.